

Evaluate a set of hypotheses with GORIC or GORICA: ANOVA Example

Rebecca M. Kuiper

24 juli 2026

Contents

Example 1: Lucas Data	1
Data preparation	1
Preparation for GORIC(A)	3
GORIC examples	4
GORICA examples	6
Example 2: Berzonsky et al.	7
Data preparation	8
Preparation for GORIC(A)	8
GORIC examples	9
GORICA examples	10
Example 3: Holubar	11
Data preparation	11
Preparation for GORIC(A)	11
Model selection using GORIC	11
Example 4: Sesame	12
Calculate GORIC values and weights	12
Calculate GORIC values and weights for H1 and its complement	14
Calculate GORIC values and weights for H2 and its complement	14
Notes on weights	15
Note 1: GORIC weights versus GORICA weights	15
Note 2: complement in case of two hypotheses of interest	15
Note 3: weights when using complement not per se higher	17
Extra: Two methods to calculate the penalty	19
Note on not full row-rank	19

Example 1: Lucas Data

Data preparation

First, load the required libraries (after they have been installed). These libraries contain functions, such as `goric`, that will be used in the R code below. Each time you reopen R, you will have to load the required libraries.

```
## First, install the packages, if you have not done this already:
if (!require("psych")) install.packages("psych")
if (!require("restriktor")) install.packages("restriktor")

## Then, load the packages:
library(psych) # for the function describeBy
library(restriktor) # for the goric function

# If you want to use restriktor from github: if (!require('pak'))
# install.packages('pak') library(pak) pak::pak('LeonardV/restriktor')
# pak::pkg_install('LeonardV/restriktor') library(restriktor) # for goric
# function
```

Second, it is necessary to load the data.

Notably, it is only possible to load the data if you are using the correct working directory (with both your R script and data file). The command `getwd()` shows you your current working directory. You can change the working directory to the one you prefer using the function `setwd()` by specifying the correct location between parentheses. Alternatively, in Rstudio, you can use the “Session” tab (on top) or you can use the “Files”-pane (on top of probably the right lower box of your Rstudio-screen, this pane is located next to the panes for “Plots”, “Packages”, “Help”, and “Viewer”).

If you open the data file `Data_Lucas.txt` in a text editor, you can see that the variable labels have been inserted (using quotes; i.e., “...”) in the first line of the file, which is called a header. Therefore, you have to specify `header = TRUE` when loading the data:

```
# Load the data
Lucas <- read.table("data/Data_Lucas.txt", header = TRUE)
```

Since a .txt file was loaded, R does not know the measurement levels of the variables and assumes all of them to be continuous, meaning that they are of interval or ratio type. Hence, especially when there are more than two groups, one has to tell R that the variable `group` is a factor by using the `factor()` function on the `group` variable (i.e., a grouping / categorical / nominal variable):

```
# Make the variable group a factor
Lucas$group <- factor(Lucas$group)
```

To inspect the first 6 rows of the dataset, use the `head()` function:

```
head(Lucas) # Look at first (6) rows of the data
```

```
  group Influence
1     1      3.58
2     1     -0.15
3     1      0.67
4     1      2.22
5     1      2.56
6     1      1.70
```

To see a more detailed overview of the data via descriptive statistics split by `group` variable, use the `describeBy()` function with `Lucas$group` set to be a grouping variable, as follows:

```
descrstat <- describeBy(Lucas$Influence, Lucas$group, mat = TRUE, digits = 3)
descrstat
```

	item	group1	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X11	1	1	1	30	2.329	1.860	2.330	2.243	2.009	-0.45	6.74	7.19	0.371	-0.719	0.340
X12	2	2	1	30	1.328	1.149	1.320	1.272	1.231	-0.38	3.87	4.25	0.318	-0.943	0.210
X13	3	3	1	30	3.200	1.790	3.475	3.243	1.824	-0.31	6.84	7.15	-0.149	-0.556	0.327

X14	4	4	1	30	2.231	1.450	1.665	2.099	1.171	0.47	5.12	4.65	0.724	-0.867	0.265
X15	5	5	1	30	3.229	1.500	3.730	3.358	1.460	-0.46	5.67	6.13	-0.630	-0.563	0.274

Preparation for GORIC(A)

ANOVA model

First, an R-object with unconstrained estimates is needed, that is, in this example, the five group means and one residual variance. The linear regression model using `lm()` function is specified as follows:

```
lm_fit_Lucas <- lm(Influence ~ group - 1, data = Lucas)
```

Note that:

1. `y ~ group - 1` instructs the function `lm` (linear model) to regress the variable `y` on the variable `group`.
2. The `- 1` instructs the function `lm` to drop the intercept and, therefore, estimate the means of each group, resulting, here, in five group means.
If the intercept is not dropped, '`y ~ group`' would estimate an intercept – representing the mean of the reference group – and the mean differences between the other (here, four) groups and the reference group.
Note that if you would have multiple factors, you may want to leave the intercept in to improve the interpretation of the parameters; which aids in specifying your hypotheses.
3. The results are collected in, what is called, an R-object; here, named `lm_fit_Lucas`.

It can be helpful to check the names used in this model, because these are needed when specifying the hypotheses:

```
names(coef(lm_fit_Lucas))
```

```
[1] "group1" "group2" "group3" "group4" "group5"
```

Note: `lm` object or `coef()` and `vcov()` When calculating the GORIC, the `goric()` function can use `lm` objects as input.

When calculating the GORICA, the `goric()` function can use `lm`, `glm`, `glmer`, `nlmer`, `lmer`, and `lavaan` objects as input, but one can also enter the (structural) parameter estimates and their covariance matrix yourself. For most R-objects, the latter can be obtained using `coef()` and `vcov()`; e.g.:

```
est <- coef(lm_fit_Lucas)
VCOV <- vcov(lm_fit_Lucas)
```

This will be illustrated in an GORICA example below.

Set of hypotheses

To evaluate the hypotheses of interest, it is necessary to specify the restrictions in these hypotheses correctly:

- Within the `goric()` function, it is possible to use the following operators: `>`, `<`, `=`, `<=`, `>=`, `==` (where the last three denote the same constraint as the first three).
- The `goric()` function can deal with:
 - pairwise restrictions separated by a semicolon ; (e.g., "`beta1 > beta2; beta2 > beta3`") or an `&` sign (e.g., "`beta1 > beta2 & beta2 > beta3`").
 - combined restrictions consisting of more than one operator (e.g., "`beta1 > beta2 > beta3`").

Note that one should use the labels of the parameter estimates (in the example above: `group1` – `group5`).

- One can also define hypothesis in terms of linear functions of parameters. For more details, see ‘Extra possibility specification hypotheses’ near the end of the `goric` tutorial called ‘Tutorial_GORIC_restriktor_General’ (<https://github.com/rebeccakuiper/Tutorials>).

Let us specify the following theory-based hypotheses:

```
H1 <- "group5 = group3 > (group1, group4) > group2"
# which equals: H1 <- 'group5 = group3 > group1 > group2; group3 > group4 >
# group2' Note: H1 is not full row-rank (see below and the goric() tutorial for
# more details).
H2 <- "group3 > group1 > group4 = group5 > group2"
```

To prevent from selecting a weak hypothesis, that is, a hypothesis not supported by the data, one should include a failsafe/safeguard hypothesis. This can be:

- comparison = “unconstrained”: This includes all possible hypotheses, thus including the one(s) of interest, which is the default in case of multiple theory-based hypotheses;
- comparison = “complement”: This includes all other possible hypotheses, thus excluding the one(s) of interest, which is the default in case of one multiple theory-based hypothesis;
- comparison = “none”: In this case, no failsafe is used.

The third option should only be used when you know that all the hypotheses in the set cover all possibilities. Namely, only in that case, (at least) one of the hypotheses contains the truth.

The first should be used when the interest lies in comparing the support for multiple theory-based hypotheses (and is therefore the default when multiple hypotheses are entered). One should then first check whether at least one theory-based hypothesis has more support than the unconstrained. If so, the non-weak hypotheses can be compared to all the other (weak and non-weak) theory-based hypotheses in the set. If not, then all the theory-based hypotheses are weak.

The second, that is, the complement, should be used in case there is one theory-based hypothesis (which is therefore the default when one theory-based hypothesis is entered). The complement (reflecting all the other possibilities) is then the competing hypothesis.

Seed values

In the calculation of the $GORIC(A)$, an iterative process is used to calculate the penalty / complexity part. Therefore, one needs to set a seed value using `set.seed()`. This has two advantages:

1. Using the same seed value leads to the same penalty value every time this code is run. This is thus helpful to reproduce results.
2. Using different seed values, allows for a sensitivity check on the penalty value. If it is sensitive, then increase number of iterations used in calculation of the penalty (see below).

GORIC examples

The GORIC can be used for normal linear models, like ANOVA and regression models. The `goric()` function calculates the $GORIC$ value by default (type = “goric”). To calculate the $GORICA$ values, the argument `type` has to be set to `gorica` (type = “gorica”, see example below).

Note: For (more) information regarding interpreting the $GORIC(A)$ output, see ‘Guidelines_output_GORIC’ (<https://github.com/rebeccakuiper/Tutorials>).

Example 1.1a: Using the unconstrained as failsafe

```
set.seed(123) # Set seed value
output <- goric(lm_fit_Lucas, hypotheses = list(H1 = H1, H2 = H2))
```

```
# summary(output)
output
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights	goric.weights
1	H1	-278.051	3.458	563.019	0.493	0.407	0.899	
2	H2	-281.761	3.136	569.794	0.012	0.561	0.030	
3	unconstrained	-278.048	6.000	568.097	0.495	0.032	0.071	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H1' is 29.586 times more supported than 'H2'.

```
round(output$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. unconstrained
H1	1.00	29.59	12.67
H2	0.03	1.00	0.43
unconstrained	0.08	2.34	1.00

It can be seen that the order-restricted hypothesis H_1 has > 1 times more support than H_u (the unconstrained hypothesis). Hence, H_1 is not a weak hypotheses and can thus be compared to the other (weak and non-weak) competing hypotheses:

H_1 is 29.6 times more likely than H_2 .

Example 1.1b: Using the complement as failsafe

```
set.seed(123) # Set seed value
output_c <- goric(lm_fit_Lucas, hypotheses = list(H1))
# summary(output_c)
output_c
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1	-278.051	3.458	563.019	0.499	0.903	0.903
2	complement	-278.048	5.691	567.479	0.501	0.097	0.097

Conclusion:

The order-restricted hypothesis 'H1' has 9.30 times more support than its complement.

The order-restricted hypothesis H_1 has 9.3 times more support than its complement.

Note: The log-likelihood (loglik) weights seem to be quite close. This could indicate that one or more of the inequality constraints can be replaced by (about-)equality constraints. One could investigate with the benchmarks function, using 'output_type = "rlw"', whether the loglik weights indeed are close.

For more information, see the guidelines ('Guidelines_output_GORIC.html') and/or the benchmark tutorial on <https://github.com/rebeccakuiper/Tutorials>.

GORICA examples

The *GORICA* can be used for a broad range of models. Besides normal linear models (e.g., ANOVA and regression models) it can be applied also to logistic regression and SEM models and much more. To calculate the *GORICA* values, one should use `type = "gorica"` in case of using an 'lm' object as input (in the other instances, "gorica" is the default).

Example 1.2a: Using the unconstrained as failsafe

```
set.seed(123) # Set seed value
output_gorica <- goric(lm_fit_Lucas, hypotheses = list(H1 = H1, H2 = H2), type = "gorica")
# summary(output_gorica)
output_gorica
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights	gorica.weights
1	H1	1.732	2.458	1.453	0.494	0.407	0.901	
2	H2	-2.071	2.136	8.414	0.011	0.561	0.028	
3	unconstrained	1.735	5.000	6.531	0.495	0.032	0.071	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H1' is 32.483 times more supported than 'H2'.

```
round(output_gorica$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. unconstrained
H1	1.00	32.48	12.67
H2	0.03	1.00	0.39
unconstrained	0.08	2.56	1.00

In Example 1, the same analysis is done with the GORIC, you can see that the (relative) weights are (about) the same for the GORIC and GORICA. The ratio of weights may differ (somewhat see a note at the end of this tutorial).

From this output, it can be seen that the order-restricted hypothesis H_1 has > 1 times more support than H_u (the unconstrained hypothesis). Hence, H_1 is not a weak hypotheses and can thus be compared to the other (weak and non-weak) competing hypotheses:

H_1 is 32.5 times more likely than H_2 .

```
set.seed(123) # Set seed value
est <- coef(lm_fit_Lucas)
VCOV <- vcov(lm_fit_Lucas)
output_gorica_alt <- goric(est, VCOV = VCOV, hypotheses = list(H1 = H1, H2 = H2),
  type = "gorica")
# summary(output_gorica_alt)
output_gorica_alt
```

Alternative input

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights	gorica.weight
1	H1	1.647	2.458	1.622	0.493	0.407	0.898	
2	H2	-2.029	2.136	8.330	0.012	0.561	0.031	
3	unconstrained	1.650	5.000	6.700	0.494	0.032	0.071	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
- * 'H1' is 28.615 times more supported than 'H2'.

```
round(output_gorica_alt$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. unconstrained
H1	1.00	28.62	12.67
H2	0.03	1.00	0.44
unconstrained	0.08	2.26	1.00

Example 1.2b: Using the complement as failsafe

```
H1 <- "group5 = group3 > (group1, group4) > group2"
# Note: H1 is not full row-rank; for more details, see below and/or the gorica
# tutorial.

set.seed(123) # Set seed value
output_gorica_c <- gorica(lm_fit_Lucas, hypotheses = list(H1), type = "gorica")
# summary(output_gorica_c)
output_gorica_c
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights
1	H1	1.732	2.458	1.453	0.499	0.903	0.903
2	complement	1.735	4.691	5.912	0.501	0.097	0.097

Conclusion:

The order-restricted hypothesis 'H1' has 9.30 times more support than its complement.

The order-restricted hypothesis H_1 has 9.3 times more support than its complement. Notably, the weights are also now the same as for the GORIC (Example 2).

Note: The log-likelihood (loglik) weights seem to be quite close. This could indicate that one or more of the inequality constraints can be replaced by (about-)equality constraints. One could investigate with the benchmarks function, using 'output_type = "rlw"', whether the loglik weights indeed are close.

For more information, see the guidelines ('Guidelines_output_GORIC.html') and/or the benchmark tutorial on <https://github.com/rebeccakuiper/Tutorials>.

Example 2: Berzonsky et al.

Next, the R code to apply the GORIC(A) to the data of Berzonsky et al. is shown. The instruction on loading and preparing the data are similar to the ones from Lucas example (Example 1). Hence, for more comments and details, see that example.

Data preparation

```
# Read Data.
BerzEtAl <- read.table("data/Data_BerzEtAl.txt", header = TRUE)
BerzEtAl$group <- factor(BerzEtAl$group)
# this command tells R that group is a factor and not a continuous variable

# Inspect data
head(BerzEtAl)
```

```
  group Influence
1     1  39.79709
2     1  26.58804
3     1  36.04999
4     1  35.92915
5     1  27.02636
6     1  31.08900
```

```
# Compute descriptive statistics for each group
descrip <- describeBy(BerzEtAl$Influence, BerzEtAl$group, mat = TRUE, digits = 3)
descrip
```

	item	group1	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X11	1	1	1	15	32.00	4.461	33.508	31.934	3.768	25.055	39.797	14.742	-0.114	-1.353	1.152
X12	2	2	1	15	21.00	4.627	21.426	20.976	3.795	13.179	29.138	15.959	-0.081	-1.007	1.195
X13	3	3	1	15	7.00	4.192	6.364	6.853	3.498	1.456	14.453	12.996	0.440	-1.100	1.082
X14	4	4	1	15	14.00	6.242	14.708	14.259	7.264	0.906	23.724	22.818	-0.349	-0.809	1.612
X15	5	5	1	15	14.00	4.368	14.068	14.195	3.532	5.194	20.266	15.072	-0.465	-0.746	1.128
X16	6	6	1	15	0.01	0.010	0.011	0.009	0.010	-0.004	0.038	0.042	1.081	1.119	0.003
X17	7	7	1	15	0.10	0.093	0.083	0.096	0.129	-0.026	0.284	0.310	0.179	-1.232	0.024
X18	8	8	1	15	0.22	0.217	0.263	0.225	0.218	-0.150	0.528	0.677	-0.491	-1.087	0.056

Preparation for GORIC(A)

```
# Using the R package lm
lm_fit_BerzEtAl <- lm(Influence ~ group - 1, data = BerzEtAl)

# Check names used in model
names(coef(lm_fit_BerzEtAl))
```

```
[1] "group1" "group2" "group3" "group4" "group5" "group6" "group7" "group8"
```

```
# Specify restrictions using those names
```

```
# Set of hypotheses
H1 <- "group1 > group2; group1 > group3; group1 > group4;
group5 > group6; group5 > group7; group5 > group8;
group1 > group5; group2 > group6; group3 > group7; group4 > group8;
group1 - group5 > group2 - group6;
group1 - group5 > group3 - group7;
group1 - group5 > group4 - group8;"

H2 <- "group1 > group2; group1 > group3; group1 > group4;
group5 > group6; group5 > group7; group5 > group8;
group1 > group5; group2 > group6; group3 > group7; group4 > group8;
group1 - group5 > 2*(group2 - group6);
```



```
group1 - group5 > 2*(group3 - group7);
group1 - group5 > 2*(group4 - group8);"
```

GORIC examples

Example 2.1a: Using the unconstrained as failsafe

```
set.seed(123) # Set seed value
output_B <- goric(lm_fit_BerzEtAl, hypotheses = list(H1 = H1, H2 = H2))
# summary(output_B)
output_B
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights	goric.weights
1	H1	-328.302	6.089	668.784	0.229	0.487	0.845	
2	H2	-353.318	6.090	718.815	0.000	0.487	0.000	
3	unconstrained	-327.087	9.000	672.173	0.771	0.027	0.155	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
* 'H1' is 7.314e+10 times more supported than 'H2'.

```
round(output_B$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. unconstrained
H1	1.00	73135417616	5.45
H2	0.00	1	0.00
unconstrained	0.18	13430577836	1.00

The order-restricted hypothesis H_1 has > 1 times more support than H_u (unconstrained). Hence, H_1 is not a weak hypotheses and can thus be compared to the other (weak and non-weak) competing hypotheses: H_1 is much more supported than H_2 .

Example 2.1b: Using the complement as failsafe

```
set.seed(123)
output_B_c <- goric(lm_fit_BerzEtAl, hypotheses = list(H1))
# summary(output_B_c)
output_B_c
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1	-328.302	6.089	668.784	0.229	0.942	0.827
2	complement	-327.087	8.869	671.911	0.771	0.058	0.173

Conclusion:

The order-restricted hypothesis 'H1' has 4.78 times more support than its complement.

The order-restricted hypothesis H_1 has 4.8 times more support than its complement.

GORICA examples

Example 2.2a: Using the unconstrained as failsafe

```
set.seed(123) # Set seed value
output_B_gorica <- gorica(lm_fit_BerzEtAl, hypotheses = list(H1 = H1, H2 = H2), type = "gorica")
# summary(output_B_gorica)
output_B_gorica
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights	gorica.weights
1	H1	-8.202	5.089	26.582	0.226	0.487	0.843	
2	H2	-39.874	5.090	89.927	0.000	0.487	0.000	
3	unconstrained	-6.974	8.000	29.947	0.774	0.027	0.157	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H1' is 5.691e+13 times more supported than 'H2'.

```
round(output_B_gorica$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. unconstrained
H1	1.00	5.691241e+13	5.38
H2	0.00	1.000000e+00	0.00
unconstrained	0.19	1.058182e+13	1.00

The order-restricted hypothesis H_1 has > 1 times more support than H_u (unconstrained). Hence, H_1 is not a weak hypotheses and can thus be compared to the other (weak and non-weak) competing hypotheses: H_1 is much more supported than H_2 .

Example 2.2b: Using the unconstrained as failsafe

```
set.seed(123)
output_B_gorica_c <- gorica(lm_fit_BerzEtAl, hypotheses = list(H1), type = "gorica")
# summary(output_B_gorica_c)
output_B_gorica_c
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights
1	H1	-8.202	5.089	26.582	0.226	0.942	0.825
2	complement	-6.974	7.869	29.685	0.774	0.058	0.175

Conclusion:

The order-restricted hypothesis 'H1' has 4.72 times more support than its complement.

The order-restricted hypothesis H_1 has 4.7 times more support than its complement.

Example 3: Holubar

Data preparation

First, read in the Holubar dataset, and tell R that the variable `gr` (group) is a factor instead of a continuous variable (although it is not necessary because it consists of only two groups).

```
Holubar <- read.table("data/Data_Holubar.txt", header = TRUE) # load the data
Holubar$gr <- factor(Holubar$gr) # tell R that gr is a factor
```

If you want a more detailed overview of the data, also by means of descriptive statistics splitted by `group`, use

```
head(Holubar)
```

```
      at gr
1 0.5549239 1
2 3.6167880 1
3 0.8071903 1
4 1.2733173 1
5 2.3898220 1
6 0.1910118 1
```

```
descrstat <- describeBy(Holubar$at, Holubar$gr, mat = TRUE, digits = 3)
descrstat
```

	item	group	1	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X11	1	1	1	20	0.98	1.20	1.177	0.930	1.118	-0.719	3.617	4.335	0.190	-0.778	0.268	
X12	2	2	1	27	0.02	1.88	0.169	0.111	1.951	-4.961	2.921	7.883	-0.517	-0.104	0.362	
X13	3	3	1	28	0.27	1.72	-0.099	0.203	2.435	-2.276	3.796	6.073	0.292	-1.025	0.325	

Preparation for GORIC(A)

ANOVA model

Then, fit an ANOVA-model by means of the `lm()` function (linear model) and directly check the names that are used in this model:

```
lm_fit_Holubar <- lm(at ~ gr - 1, data = Holubar)
names(coef(lm_fit_Holubar))
```

```
[1] "gr1" "gr2" "gr3"
```

Set of hypotheses

The following hypothesis will be evaluated:

```
H1 <- "gr2 > gr1 > gr3"
```

Model selection using GORIC

Calculate the GORIC values and weights:

```
set.seed(123) # Set seed value
output_Hol <- goric(lm_fit_Holubar, hypotheses = list(H1))
# summary(output_Hol)
output_Hol
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1	-144.981	2.803	295.569	0.125	0.710	0.259
2	complement	-143.038	3.697	293.469	0.875	0.290	0.741

Conclusion:

The order-restricted hypothesis 'H1' has less support (namely, $0.259 / 0.741 < 1$ times more support) than the complement hypothesis.

Example 4: Sesame

```
# read in the sesame data from a text file
Sesame <- read.table("data/sesamesimANOVA.txt", header = TRUE)

# make viewcat a factor, that is, a categorical variable
Sesame$viewcat <- factor(Sesame$viewcat)

# Inspect data
head(Sesame)
```

```
  viewcat postnumb
1      1 14.51050
2      3 35.30549
3      3 22.68280
4      1 39.82436
5      4 40.42370
6      3 22.87055
```

```
# estimate the parameters of the statistical model at hand
fit_Sesame <- lm(postnumb ~ viewcat - 1, data = Sesame)

# Check names used in model
names(coef(fit_Sesame))
```

```
[1] "viewcat1" "viewcat2" "viewcat3" "viewcat4"
```

```
# Specify restrictions using those names
```

```
# Example hypotheses:
H1 <- "viewcat1 = viewcat2 < viewcat3 < viewcat4"
H2 <- "viewcat1 < viewcat2 < viewcat3 < viewcat4"
H3 <- "viewcat1 = viewcat2 < viewcat3 = viewcat4"
```

Calculate GORIC values and weights

Here, we assume that the interest lies in H1 to H3. Since these do not cover all possibilities, the unconstrained hypotheses is included in the set.

```
set.seed(123)
goric_sesam <- goric(fit_Sesame, hypotheses = list(H1, H2, H3))
# summary(goric_sesam)
goric_sesam
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights	goric.weights
-------	--------	---------	-------	----------------	-----------------	---------------	---------------

1	H1	-918.485	2.810	1842.590	0.065	0.310	0.134
2	H2	-916.540	3.094	1839.269	0.457	0.233	0.704
3	H3	-919.644	2.500	1844.287	0.021	0.422	0.057
4	unconstrained	-916.540	5.000	1843.080	0.457	0.035	0.105

Conclusion:

- The order-restricted hypothesis 'H2' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H2' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H2' is 5.262 times more supported than 'H1'.
 - * 'H2' is 12.293 times more supported than 'H3'.

```
round(gorica_sesam$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. H3	vs. unconstrained
H1	1.00	0.19	2.34	1.28
H2	5.26	1.00	12.29	6.72
H3	0.43	0.08	1.00	0.55
unconstrained	0.78	0.15	1.83	1.00

From the output, it is concluded that H2 is not a weak hypothesis (nor is H1). Thus, its support can be compared to that of the other hypotheses:

H2 is 5.3 times more supported than H1 and 12.3 times more than H3.

Hence, H2 is the preferred hypothesis (and has quite some evidence).

Calculate GORICA values and weights

```
set.seed(123)
gorica_sesam <- gorica(fit_Sesame, hypotheses = list(H1, H2, H3), type = "gorica")
# summary(gorica_sesam)
gorica_sesam
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights	gorica.weights
1	H1	-7.051	1.810	17.723	0.064	0.310	0.132	
2	H2	-5.091	2.094	14.371	0.458	0.233	0.707	
3	H3	-8.235	1.500	19.470	0.020	0.422	0.055	
4	unconstrained	-5.091	4.000	18.182	0.458	0.035	0.105	

Conclusion:

- The order-restricted hypothesis 'H2' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H2' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H2' is 5.346 times more supported than 'H1'.
 - * 'H2' is 12.801 times more supported than 'H3'.

```
round(gorica_sesam$ratio.gw, 2)
```

	vs. H1	vs. H2	vs. H3	vs. unconstrained
H1	1.00	0.19	2.39	1.26
H2	5.35	1.00	12.80	6.72
H3	0.42	0.08	1.00	0.53
unconstrained	0.80	0.15	1.90	1.00

Calculate GORIC values and weights for H1 and its complement

Here, we assume that the interest lies only in H1. In that case, it should be evaluated against its complement.

```
set.seed(123)
goric_sesam_1c <- goric(fit_Sesame, hypotheses = list(H1))
# summary(goric_sesam_1c)
goric_sesam_1c
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1	-918.485	2.810	1842.590	0.125	0.868	0.484
2	complement	-916.540	4.690	1842.460	0.875	0.132	0.516

Conclusion:

The order-restricted hypothesis 'H1' has less support (namely, $0.484 / 0.516 < 1$ times more support) than its complement.

From the output, it is concluded that the support for H1 is comparable to that of its complement, with a slight preference for the complement. Note that the complement has the highest (log) likelihood but also the highest complexity/penalty, when balancing fit and complexity this results in a slight preference for the complement of H1.

Calculate GORICA values and weights for H1 and its complement

```
set.seed(123)
gorica_sesam_1c <- goric(fit_Sesame, hypotheses = list(H1), type = "gorica")
# summary(gorica_sesam_1c)
gorica_sesam_1c
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights
1	H1	-7.051	1.810	17.723	0.123	0.868	0.480
2	complement	-5.091	3.690	17.561	0.877	0.132	0.520

Conclusion:

The order-restricted hypothesis 'H1' has less support (namely, $0.480 / 0.520 < 1$ times more support) than its complement.

Calculate GORIC values and weights for H2 and its complement

Here, we assume that the interest lies only in H2. In that case, it should be evaluated against its complement.

```
set.seed(123)
goric_sesam_2c <- goric(fit_Sesame, hypotheses = list(H2 = H2))
# summary(goric_sesam_2c)
goric_sesam_2c
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H2	-916.540	3.094	1839.269	0.765	0.855	0.950
2	complement	-917.718	4.870	1845.176	0.235	0.145	0.050

Conclusion:

The order-restricted hypothesis 'H2' has 19.17 times more support than its complement.

From the output, it is concluded that H1 is about 19 times more likely than its complement, showing quite some support for H2.

Calculate GORICA values and weights for H2 and its complement

```
set.seed(123)
gorica_sesam_2c <- goric(fit_Sesame, hypotheses = list(H2 = H2), type = "gorica")
# summary(gorica_sesam_2c)
gorica_sesam_2c
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights
1	H2	-5.091	2.094	14.371	0.766	0.855	0.951
2	complement	-6.274	3.870	20.289	0.234	0.145	0.049

Conclusion:

The order-restricted hypothesis 'H2' has 19.28 times more support than its complement.

Notes on weights

Here, the example of Lucas is used again; thus, using the following hypotheses of interest:

```
H1 <- "group5 = group3 > (group1, group4) > group2"
# Note: H1 is not full row-rank; for more details, see below and/or the goric
# tutorial.
H2 <- "group3 > group1 > group4 = group5 > group2"
```

Note 1: GORIC weights versus GORICA weights

The GORICA weights (asymptotically) equal the GORIC weights. The differences are minor and often not notable with 2 decimals. Because of these minor differences, the relative weights (i.e., ratio of weights) can differ. Those differences in relative weights can even be large (as is in the Berzonsky et al Example), when dividing a very large number by a very small number with minor differences in these values.

Note 2: complement in case of two hypotheses of interest

One cannot compare the support of hypotheses when comparing them to their complements. Instead, one should evaluate them simultaneously in one set (like done above):

```
# Calculate goric for H1 and its complement
set.seed(123) # Set seed value
output_c_H1 <- goric(lm_fit_Lucas, hypotheses = list(H1))
# summary(output_c_H1)
output_c_H1
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1	-278.051	3.458	563.019	0.499	0.903	0.903

2	complement	-278.048	5.691	567.479	0.501	0.097	0.097
---	------------	----------	-------	---------	-------	-------	-------

Conclusion:

The order-restricted hypothesis 'H1' has 9.30 times more support than its complement.

```
# Calculate goric for H2 and its complement
set.seed(123) # Set seed value
output_c_H2 <- goric(lm_fit_Lucas, hypotheses = list(H2 = H2))
# summary(output_c_H2)
output_c_H2
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H2	-281.761	3.136	569.794	0.024	0.938	0.270
2	complement	-278.048	5.853	567.803	0.976	0.062	0.730

Conclusion:

The order-restricted hypothesis 'H2' has less support (namely, $0.270 / 0.730 < 1$ times more support) than its complement.

```
# Calculate goric for H1 and H2 (and Hu):
set.seed(123) # Set seed value
output_H1H2 <- goric(lm_fit_Lucas, hypotheses = list(H1, H2)) # Note: by default,
# against the unconstrained summary(output_H1H2)
output_H1H2
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights	goric.weights
1	H1	-278.051	3.458	563.019	0.493	0.407	0.899	
2	H2	-281.761	3.136	569.794	0.012	0.561	0.030	
3	unconstrained	-278.048	6.000	568.097	0.495	0.032	0.071	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H1' is 29.586 times more supported than 'H2'.

```
output_H1H2$ratio.gw[1, 2]
```

```
[1] 29.58579
```

```
# The latter is not equal to:
```

```
output_c_H1$ratio.gw[1, 2]/output_c_H2$ratio.gw[1, 2]
```

```
[1] 25.15443
```

Extra

```
# Notably, you could derive the support from H1 vs H2 from their support versus
# that of Hu:
set.seed(123) # Set seed value
output_u_H1 <- goric(lm_fit_Lucas, hypotheses = list(H1), comparison = "unconstrained")
# summary(output_u_H1)
output_u_H1
```


restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1	-278.051	3.458	563.019	0.499	0.927	0.927
2	unconstrained	-278.048	6.000	568.097	0.501	0.073	0.073

Advise: Are you certain you wish to assess the order-restricted hypothesis in comparison to the unconstrained?

Conclusion:

The order-restricted hypothesis 'H1' has $0.927 / 0.073 > 1$ times more support than the unconstrained.

```
#
set.seed(123) # Set seed value
output_u_H2 <- goric(lm_fit_Lucas, hypotheses = list(H2 = H2), comparison = "unconstrained")
# summary(output_u_H2)
output_u_H2
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H2	-281.761	3.136	569.794	0.024	0.946	0.300
2	unconstrained	-278.048	6.000	568.097	0.976	0.054	0.700

Advise: Are you certain you wish to assess the order-restricted hypothesis in comparison to the unconstrained?

Conclusion:

The order-restricted hypothesis 'H2' has less support (namely, $0.300 / 0.700 < 1$ times more support) than the unconstrained.

```
#
output_u_H1$ratio.gw[1, 2]/output_u_H2$ratio.gw[1, 2]
```

```
[1] 29.58579
```

```
# which (approximately) equals:
output_H1H2$ratio.gw[1, 2]
```

```
[1] 29.58579
```

Note 3: weights when using complement not per se higher

In the Lucas example:

- The order-restricted hypothesis H_1 has 12.7 times more support than H_u (unconstrained).
- The order-restricted hypothesis H_1 has 9.3 times more support than its complement.

Now, the complement does not render a higher weight than if H_u was used.

If H_m is not in agreement with the data, the complement does not always render a higher weight. This is actually a good thing: Against H_u , H_m might obtain too much support then.

Some more explanation

This is because for this particular example the mean of group 3 and 5 are close (and they are compared in Hypothesis H_1). Notably, it will hold for all of the following three hypothesis:

```
H1_gr <- "group5 > group3 > group1 > group2; group3 > group4 > group2"
# not full row-rank
H1_sm <- "group5 < group3 > group1 > group2; group3 > group4 > group2"
# not full row-rank
H1 <- "group5 = group3 > group1 > group2; group3 > group4 > group2"
# not full row-rank
```

Because the means of groups 3&5 are similar (and the other restrictions are in agreement with the data), the log likelihood values (LL's) of H_1 , H_u , and H_c (i.e., the complement of H_1) will be close - since the restricted/bounded solution (which is in agreement with H_1) is near the (unconstrained) maximum likelihood estimate (mle). Since the penalty for the complement is (always) lower than for H_u , it will receive more support than H_u - thus H_1 receives less. Notably, when H_1 is very specific (as in Berzonsky et al Example below), the penalty of H_u and H_c are almost the same.

When the means of groups 3 and 5 differ more, then evaluating against the complement does render a higher weight (than if H_u were used): Let us increase the mean of group 5 with 0.5 points (and change the name of the data & the analysis):

```
Lucas2 <- Lucas
Lucas2$Influence[Lucas$group == 5] <- Lucas$Influence[Lucas$group == 5] + 0.5
# describeBy(Lucas$Influence, Lucas$group, mat = TRUE)
# describeBy(Lucas2$Influence, Lucas$group, mat = TRUE)
lm_fit_Lucas2 <- lm(Influence ~ group - 1, data = Lucas2)
# Now, from the three hypotheses H1_gr, H1_sm, and H1, # the first (H_gr) is
# correct. Hence, that one is used to illustrate the case where # a correct
# hypothesis obtains more support when it is evaluated against its complement:
```

```
H1_gr <- "group5 > group3 > group1 > group2; group3 > group4 > group2"
# not full row-rank
set.seed(123) # Set seed value
output_u_gr <- goric(lm_fit_Lucas2, hypotheses = list(H1_gr = H1_gr), comparison = "unconstrained")
# summary(output_u_gr)
output_u_gr
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1_gr	-278.048	3.683	563.464	0.500	0.910	0.910
2	unconstrained	-278.048	6.000	568.097	0.500	0.090	0.090

Advise: Are you certain you wish to assess the order-restricted hypothesis in comparison to the unconstrained?

Conclusion:

The order-restricted hypothesis 'H1_gr' has 0.910 / 0.090 > 1 times more support than the unconstrained

*# Notably, this is the maximum support H1 can receive versus Hu (because the
log-likelihoods (LLs) are the same).*

```
set.seed(123) # Set seed value
output_c_gr <- goric(lm_fit_Lucas2, hypotheses = list(H1_gr = H1_gr))
# summary(output_c_gr)
output_c_gr
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights
1	H1_gr	-278.048	3.683	563.464	0.706	0.902	0.957
2	complement	-278.923	5.900	569.646	0.294	0.098	0.043

Conclusion:

The order-restricted hypothesis 'H1_gr' has 22.01 times more support than its complement.

Now, when the means of groups 3 and 5 differ more, then evaluating against the complement does render more support than if H_u were used: 22 versus 10.

Extra: Two methods to calculate the penalty

There are two methods that can be used in calculating the penalty. The default method is often much faster (if the number of parameters is not too high) and needs less input specification. It can, however, not deal with hypotheses that are not of full row-rank (like H_1 above). In that case, the `goric` functions uses automatically the other (bootstrap) method.

To use this bootstrap method use 'mix_weights = "boot"':

```
# if (!require('parallel')) install.packages('parallel') library(parallel)
# nrCPUcores <- detectCores(all.tests = FALSE, logical = TRUE)

set.seed(123) # Set seed value
output_b <- goric(lm_fit_Lucas, hypotheses = list(H1 = H1, H2 = H2), mix_weights = "boot")
# summary(output_b)
output_b
```

restriktor (0.7): generalized order-restricted information criterion:

Results:

	model	loglik	penalty	goric	loglik.weights	penalty.weights	goric.weights	goric.weights
1	H1	-278.051	3.460	563.022	0.493	0.407	0.899	
2	H2	-281.761	3.138	569.798	0.012	0.561	0.030	
3	unconstrained	-278.048	6.000	568.097	0.495	0.032	0.071	

Conclusion:

- The order-restricted hypothesis 'H1' is the best in the set, as it has the highest GORIC(A) weight.
- Since 'H1' has a higher GORIC(A) weight than the unconstrained hypothesis, it is not considered weak.
 - * 'H1' is 29.599 times more supported than 'H2'.

This, of course, renders the same results as above (if there is a difference, it is in the second decimal of the penalty).

Note on not full row-rank

If the restriction matrix is not of full row-rank, this means one of the following:

- a) There is at least one redundant restriction.

Then, either a.1) leave the redundant one out or a.2) use another (more time-consuming) way of obtaining the level probabilities for the penalty term (the `goric` function does this by default): Bootstrapping, as discussed above.

- b) There is at least one range restriction (e.g., $-2 < \text{group1} < 2$).

Such a restriction can be evaluated but there is a sensitivity (of a scaling factor in the covariance matrix, like with a prior in a Bayes factor) which currently cannot be checked for.

c) There is at least one conflicting restriction (e.g., $2 < \text{group1} < -2$).

Such a restriction can evidently never hold and is thus impossible to evaluate. To prevent this type of error, delete the one that is incorrect and run `goric()` again.